Assignment
Tutorial 1+ Regression analysis

Course title: Introduction to Probability and Statistics

Course code: STAT 251
2nd Year 2nd Semester Examination 2020

Submitted to-

Md. Habibur Rahman
Associate Professor
Department of Statistics
Jahangirnagar University
Savar, Dhaka

Date of Submission: February 26, 2021

Submitted by

Sl	Class Roll	Name
01	342	Tama Shil

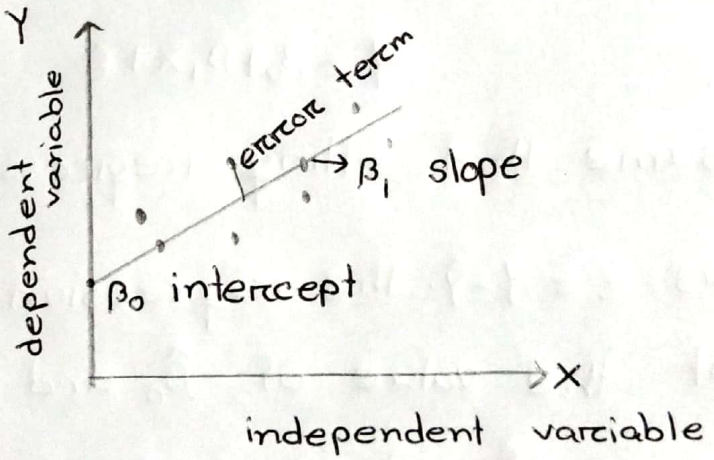
Scanned with CamScanner

"Regression"

Regression analysis is the method used for the estimation of relationship between a dependent and one or more independent variables.

Assume x and Y be some random variables and Y is dependent, x is independent. Assume X and Y follow regression model -

$$Y = \beta_0 + \beta_1 X + \varepsilon$$



Let this is a positive regression,

β_0 is the intercept term which is the parameter that measures the value of Y in absence of X.

Scanned with CamScanner

β_1 is called the regression coefficient of Y on X which measures the change of Y for a unit change in the value of X.

ϵ is called the error term.

Least square estimation is fitting a straight line regression model.

Let $(x_1, Y_1), (x_2, Y_2) \cdots (x_n, Y_n)$ be n pairs of samples for random variable X and Y. Assume X and Y follow regression model,

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Assume that fitting regression is $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x$

Thus $\epsilon = Y - \hat{Y}$. The regression analysis indicates that the value of $\hat{\beta}_0$ and $\hat{\beta}_1$ are found using n pairs of values such a way that sum of square deviation of Y and $\hat{Y}$ are minimum.

$$\sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} (Y_i - \hat{Y})^2$$

$$\sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} (Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

Scanned with CamScanner

Let $f(\hat{\beta}_0, \hat{\beta}_1) = \sum_{i=1}^{n} \varepsilon^2 = \sum_{i=1}^{n} (Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$

$\therefore f(\hat{\beta}_0, \hat{\beta}_1) = \sum_{i=1}^{n} (Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$

The value of $f(\hat{\beta}_0, \hat{\beta}_1)$ will be minimum if the values of $\hat{\beta}_0$ and $\hat{\beta}_1$ are found out such way that

$$\frac{\delta f}{\delta \hat{\beta}_0} = 0 \quad \text{and} \quad \frac{\delta f}{\delta \hat{\beta}_1} = 0$$

the two equation gives -

$$\sum_{i=1}^{n} Y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^{n} x_i$$

$$\sum_{i=1}^{n} x_i Y_i = \beta_0 \sum_{i=1}^{n} x_i + \left(\sum_{i=1}^{n} x_i^2\right) \cdot \hat{\beta}_1$$

these equation are known as normal equations. After solving these we get -

$$\boxed{\hat{\beta}_0 = \bar{Y} - \beta_1 \bar{X}}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} x_i Y_i - \frac{\sum_{i=1}^{n} Y_i \sum_{i=1}^{n} x_i}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}$$

$$= \frac{SP(XY)}{SS(X)}$$

$$= \frac{\frac{SP(XY)}{n-1}}{\frac{SS(X)}{n-1}}$$

Scanned with CamScanner

$$\hat{\beta}_1 = \frac{cov(XY)}{var(X)}$$

here $SP(XY) = \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}$

$SS(X) = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$

⇒ Sample mean, $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$

⇒ Sample variance, $s_x^2 = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \bar{x})^2$

⇨ Sample covariance,

$$Cov(XY) = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

"Applications"

- Regression is a powerful statistical technique that can generate insights on consumer behavior.
- It helps to understant business better.
- Comprehend factors influencing profitability
- It can put a service to evaluating trends and forecasting data in a variety of fields.

Scanned with CamScanner

We can use regression in our day-to-day problems related to supporting decision making, minimizing errors, increasing operational efficiency, discovering new insights and creating predictive analysis.

"Example"

x(height)	57	58	62	62	66	70	73	77	79	80	81	83
Y(weight)	66	61	71	68	63	61	68	67	79	71	76	73

Fit a regression line to estimate weight and what is the value of weight for height 75.

X	Y	X^2	Y^2	XY
57	66	3249	4356	3762
58	61	3364	3721	3538
62	71	3844	5041	4402
62	68	3844	4624	4216
66	63	4356	3969	4158
70	61	4900	3721	4270
73	68	5329	4624	4964
77	67	5929	4489	5159
80	71	6400	5041	5680
81	76	6561	5776	6156
83	73	6889	5329	6059
79	79	6241	6241	6241
$\Sigma x_i = 848$	$\Sigma y_i = 824$	$\Sigma x^2 = 60906$	$\Sigma Y^2 = 56932$	$\Sigma XY = 58605$

Scanned with CamScanner

Now, we need to get

$$Y = \hat{\beta}_0 + \hat{\beta}_1 X$$

where $Y \rightarrow$ weight

$X \rightarrow$ height

$$\hat{\beta}_1 = \frac{SP(XY)}{SS(X)}$$

$$SP(XY) = \sum_{i=1}^{n} x_i y_i - n\bar{X}\bar{Y}$$

$$\bar{X} = \frac{848}{12} = 70.7$$

$$\bar{Y} = \frac{824}{12} = 68.7$$

$$\Sigma XY = 58605$$

$$SP(XY) = 58605 - 12 \times 70.7 \times 68.7$$

$$= 319.92$$

$$SS(X) = \sum_{i=1}^{n} x_i^2 - n\bar{X}^2$$

$$= 60906 - (12 \times 70.7)$$

$$= 924.12$$

$$\hat{\beta}_1 = \frac{319.92}{924.12} = 0.35$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = 68.7 - 0.35 \times (70.7)$$

$$= 44$$

$\therefore Y = 44 + 0.35X$

weight $= 44 + 0.35$ height

If height $= 75$, weight $= 44 + 0.35 \times 75$

$= 70.25$.

Scanned with CamScanner

Tuitorcial - 01

Tama shil

342

Q1. A sample of 26 offshorce oil worckercs took parct in a simulated exercelse, rcesulting in the accompany ing data on time (sec) to complete escape:

325, 325, 334, 339, 356, 356, 359, 359, 363
364, 364, 366, 369, 370, 373, 373, 374, 375
389, 392, 393, 394, 397, 402, 403, 424

a) Construct a stem-and-leaf display of data. How does it suggest that the sample mean and median will compare?

Solution: The stem-and-leaf plot of data:

stem	leaf
32	5,5
33	4,9
35	6,6,9,9
36	3,4,4,6,9
37	0,3,3,4,5
38	9
39	2,3,4,7
40	2,3
42	4

The display is rceasonably sm symmetrcic, so the mean and median will be close.

—o—

Scanned with CamScanner

b) Calculate the values of the sample mean and median?

here, $n = \cancel{26}\ 26$ which is even,

∴ Sample mean, $\bar{x} = \frac{1}{n} \cdot \sum_{i=1}^{n} x_i$

$= \frac{1}{26}(325+325+334+339+ \text{- - - - -} +424)$

$= \frac{9638}{26}$

$= 370.692$

Median, $Me = \frac{1}{2}\left[x_{\frac{n}{2}} + x_{\frac{n}{2}+1}\right]$

$= \frac{1}{2}(x_{13} + x_{14})$

$= \frac{1}{2}(369+370)$

$= 369.5$

(c) By how much could the largest time, currently 424, be increased without affecting the value of sample median? By how much could the value be decreased without affecting the value of sample median?

Scanned with CamScanner

Solution:

Here our sample median is 369.5.

Here 369.5 < 424.

~~The last~~ So, 424 can be increased upto any number, it will not affect median.

Our median is in between 369 and 370.

So, if the number 424 is changed by (424 - 370 = 54) or more it will affect our sample median. So, 424 cannot decreased more than 54.

—o—

d) What are the values of Trimmed mean, Trimean, and IQM when observations are re-expressed in minutes? $\alpha = 10$.

Solution:

10 % trimmed mean $\bar{X}_{tr(10)}$

$n = 26$, 10% of $n = 2.6 = 3$

$$\therefore \bar{X}_{tr(10)} = (339 + 356 + 356 + 359 + 369 + 363 + 364 + 364 + 366 + 370 + 373 + 373 + 374 + 375 + 389 + 392 + 393 + 394 + 397)/20$$

$$= 371.25 \text{ sec} = 6.1875 \text{ minutes.}$$

Scanned with CamScanner

$Q_i = \frac{1}{2}\left[x_{\frac{in}{4}} + x_{\frac{in}{4}+1}\right]$ n is even

$Q_1 = \frac{1}{2}(x_6 + x_7)$

$= \frac{1}{2}(356+359)$

$= 357.5$

$Q_2 = \frac{1}{2}(x_{13}+x_{14}) = 369.5$

$Q_3 = \frac{1}{2}(x_{19}+x_{20})$

$= \frac{1}{2}(389+392)$

$= 390.5$

$\therefore TM = \frac{Q_1+2Q_2+Q_3}{4}$

$= \frac{357.5 + (2\times 369.5) + (390.5)}{4} = 371.75$ sec

$= 6.195$ minutes.

Interquartile mean,

$$\bar{X}_{IQM} = \frac{2}{n}\sum_{i=\frac{n}{4}+1}^{\frac{3n}{4}} x_i$$

$= \frac{2}{26}(359+359+363+ \cdots + 374+375+389)$

$= 369.076$ sec

$= 6.1513$ minutes.

—o—

Scanned with CamScanner

Q_2. A certain type of refrigerator comes in three sizes, each of which is available with or without an icemaker. The accompanying table gives the relevant probabilities for randomly selected purchaser:

		Size		
		17.5 ft^3	21 ft^3	24.5 ft^3
Icemaker	Yes	0.12	0.17	0.24
	No	0.18	0.13	0.16

a) what is the probability that the purchaser selected a 21-ft^3 refrigerator? A refrigerator with an icemaker?

Suppose,

A = the purchaser selected a 21 ft3 refrigerator

$\therefore P(A) = 0.17 + 0.13 = 0.30$

B = the purchaser selected a refrigerator with ice-maker

$P(B) = 0.18 + 0.017 + 0.24 = 0.53$.

—∘—

Scanned with CamScanner

(ii) Given that the purchasers select a model with ice makers, what is the probability it was a 21 ft3? How does it compare to the unconditional probability of selecting a 21 - ft3 model?

The purchasers select a model with icemaker, the probability of it was a 21 ft3

$$= \frac{0.17}{0.12+0.17+0.24}$$

$$= 0.321$$

The probability of selecting a 21 ft3 model unconditionally $= 0.17+0.13$

$= 0.30$

which is less that conditional probability.

— o —

(iii) Given the purchasers select a model with at least 20 ft3. what is the probability that the selected refrigerators didnot ~~m~~ have an icemakers?

Scanned with CamScanner

Solution:

Let A = Purchaserr selected a model with at least 20 ft3

B = the refrigeratorr did not have an ice-makerr.

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$= \frac{0.13 + 0.16}{0.13+0.17+0.24+0.16}$$

$$= 0.414.$$

—o—

Q3. An insurance company offerrs its policyholderrs a numberr of differrent prremium payment options. Forr a randomly selected policy holderr, Let X = numberr of months between successive payments. The cdf of X is as follows -

(i) What is the pmf of X?

(ii) Using just the cdf, compute $P[3 \le x \le 10]$ and $P[4 \le X]$?

$$F_x(x) = \begin{cases} 0 & x < 1 \\ 0.30 & 1 \le x < 3 \\ 0.40 & 3 \le x < 4 \\ 0.45 & 4 \le x < 6 \\ 0.60 & 6 \le x < 12 \\ 1 & 12 \le x \end{cases}$$

Scanned with CamScanner

(i) Solution

X	$F_x(x)$	$P(x)$
$x<1$	0	0
$1 \le x < 3$	0.30	0.30
$3 \le x < 4$	0.40	0.10
$4 \le x < 6$	0.45	0.5
$6 \le x < 12$	0.60	0.15
$12 \le x$	1	0.4

We use

$$P(x_i) = F(x_i) - F(x_{i-1})$$

— o —

(ii) Solution

$$P[3 \le X \le 10] = F[10] - F(2)$$

$$= 0.60 - 0.30$$

$$= 0.30$$

$$P[4 \le X] = 1 - F(3) = 1 - 0.40 = 0.60$$

— o —

Q4. Define Scatter Diagram, Rank correlation, Correlation, Regression analysis. State the characteristic of Karl Pearson's correlation coefficient. Comment of the following:

(I) $r = 0.95$ (II) $r = 0$ (III) $r = -0.90$

(IV) $r = 0.62$

Scanned with CamScanner

Scatter Diagram

A scatter Diagram also known as a scatter plot is a tool for analysis relationship between two variables for determining how closely the two variables are related.

Rank correlation

Rank correlation is the method of finding the degree of association between two variables.

Correlation and Regression analysis

Correlation analysis is the process of measuring the amount of linear relationship of two variables.

Regression is the method to estimate the relationship between a independent variable and one or more independent variables.

Scanned with CamScanner

Correlation co-efficient

The amount of linear relationship of two variables is correlation co-efficient. It is denoted by r and lies between 1 to -1.

Types of correlation coefficient

Perfect Positive correlation

If x and y have a strong positive linear correlation, r is close to +1 which indicates a perfect positive fit

Perfect negative correlation

If x and y have a strong negative linear correlation, r is close to -1 which indicates a perfect negative fit

Moderately Positive correlation → 0.30 to 0.70

$0.30 < r < 0.70$

Moderate negative correlation →

$-0.30 < r < -0.70$ -0.30 to -0.7

weak positive ~~lin~~ correlation → 0 to 0.3

$0 < r < 0.3$

Scanned with CamScanner

Moderate Positive correlation → 0.3 to 0.5

Weak Positive correlation → 0 to 0.3

Strong positive correlation → 0.5 to 0.7

very strong positive correlation → 0.7 to 1

Weak negative correlation → 0 to -.3

Moderate negative correlation → -0.3 to -0.5

A strong negative correlation → -0.5 to -0.7

very strong negative correlation → -0.7 to -1

No correlation → r is close to 0.

(i) $r = 0.95$

as r is between 0.7 to 1. It is a very strong positive correlation.

(ii) $r = 0$

as r is zero, there is no correlation

(iii) $r = -0.90$

as r is between -0.7 to -1, it is a very strong negative correlation

(iv) $r = 0.62$

As r is between 0.5 to 0.7, it is a strong positive correlation.

Scanned with CamScanner

Q5. A bank operate both drive up facility and a walk-up window. On a randomly selected day X = the proportion of time that the drive up facility is in use (at least one customer is being served or waiting to be served) and Y = the proportion of time that the walk up window is in use. Then the set of possible values for (X,Y) is the rectangle $D = \{(x,y): 0 \le x \le 1;\ 0 \le y \le 1\}$. Suppose, the joint density function of (X,Y) is given by - $f_{XY}(x,y) = \frac{6}{5}(x+y^2);\ 0 \le x \le 1,\ 0 \le y \le 1$

(i) Verify that $f_{XY}(x,y)$ is a legitimate pdf?

Given. $f_{XY}(x,y) = \frac{6}{5}(x+y^2) \quad 0 \le x \le 1,\ 0 \le y \le 1$

Now,

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy$$

$$= \int_0^1 \int_0^1 \frac{6}{5}(x+y^2)\,dx\,dy$$

$$= \int_0^1 \frac{6}{5}\left[\frac{x^2}{2} + xy^2\right]_0^1 dy$$

$$= \frac{6}{5}\int_0^1 \left(\frac{1}{2} + y^2\right) dy$$

Scanned with CamScanner

$$= \frac{6}{5}\left[\frac{y}{2} + \frac{y^3}{3}\right]_0^1$$

$$= \frac{6}{5}\left(\frac{1}{2} + \frac{1}{3}\right)$$

$$= \frac{6}{5} \times \frac{5}{6}$$

$$= 1$$

$\therefore f_{XY}(x,y)$ is a legitimate pdf.

—o—

(ii) What is the probability that neither facility is busy more than one quarter of time?

$$P\left(x \le \frac{1}{4}, y \le \frac{1}{4}\right)$$

$$= \int_0^{1/4}\int_0^{1/4} f_{XY}(x,y)\,dx\,dy$$

$$= \int_0^{1/4}\int_0^{1/4} \frac{6}{5}(x+y^2)\,dx\,dy$$

$$= \frac{6}{5}\int_0^{1/4}\left[\frac{x^2}{2} + xy^2\right]_0^{1/4} dy$$

$$= \frac{6}{5}\int_0^{1/4}\left(\frac{1}{32} + \frac{1}{4}y^2\right) dy$$

$$= \frac{6}{5}\left[\frac{y}{32} + \frac{y^3}{12}\right]_0^{1/4}$$

$$= \frac{6}{5}\left(\frac{1}{32\times 4} + \frac{1}{12\times 4^3}\right)$$

$$= \frac{6}{5} \times \left(\frac{1}{128} + \frac{1}{768}\right)$$

$$= \frac{7}{640}.$$

—o—

Scanned with CamScanner

(iii) Find the marginal pdf of X and marginal pdf of Y.

Marginal pdf of X,

$$f_x(x) = g(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$$

$$= \int_0^1 \frac{6}{5}(x+y^2)\,dy$$

$$= \frac{6}{5}\left[xy + \frac{y^3}{3}\right]_0^1$$

$$= \frac{6}{5}\left(x + \frac{1}{3}\right)$$

$$= \frac{6x}{5} + \frac{2}{5}, \quad 0 \le x \le 1$$

Marginal pdf of Y,

$$f_y(y) = h(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$$

$$= \int_0^1 \frac{6}{5}(x+y^2)\,dx$$

$$= \frac{6}{5}\left[\frac{x^2}{2} + xy^2\right]_0^1$$

$$= \frac{6}{5}\left(\frac{1}{2} + y^2\right)$$

$$= \frac{3}{5} + \frac{6y^2}{5}; \quad 0 \le y \le 1.$$

—o—

Scanned with CamScanner